

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 22: Heights and Distances

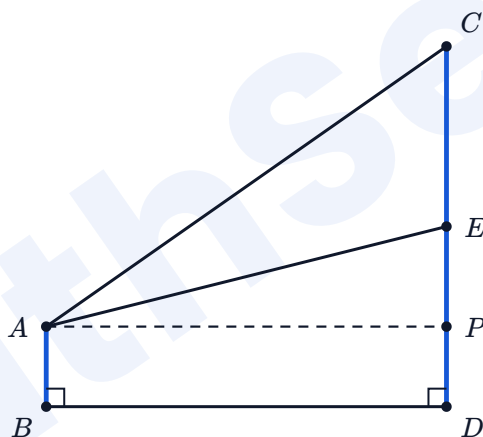
Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Problems with two triangles

1. A vertical tower AB stands 100 m from the foot D of a building CD . E is the mid-point of CD . The angle of depression of the top A of the tower is 35° from C and 14° from E . Using tangent values rounded to two decimal places, find the height of (a) the tower (b) the building.

ICSE 2025, 5 marks



SOLUTION

Draw AP horizontal, meeting CD at P . Then $AP = BD = 100$ m and $PD = AB$. The angles of depression equal $\angle CAP = 35^\circ$ and $\angle EAP = 14^\circ$ (alternate angles).

From the tables, $\tan 35^\circ = 0.70$ and $\tan 14^\circ = 0.25$.

$CP = 100 \tan 35^\circ = 70$ m and $EP = 100 \tan 14^\circ = 25$ m, so $CE = 70 - 25 = 45$ m.

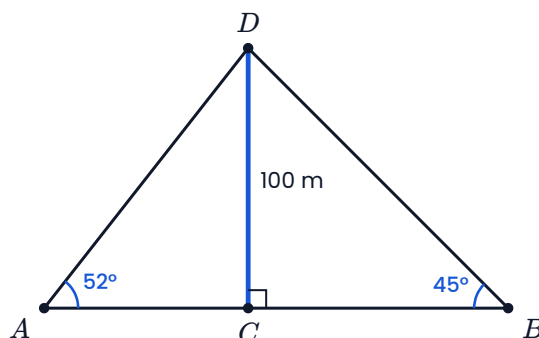
E is the mid-point of CD , so $CD = 2 \times 45 = 90$ m and $ED = 45$ m.

$AB = PD = ED - EP = 45 - 25 = 20$ m

Answer: (a) tower 20 m (b) building 90 m

2. A tree CD is 100 m high. From two points A and B on the ground on opposite sides of the tree, its top D is seen at angles of elevation of 52° and 45° . Find the distance AB to the nearest metre.

ICSE 2024, 5 marks



SOLUTION

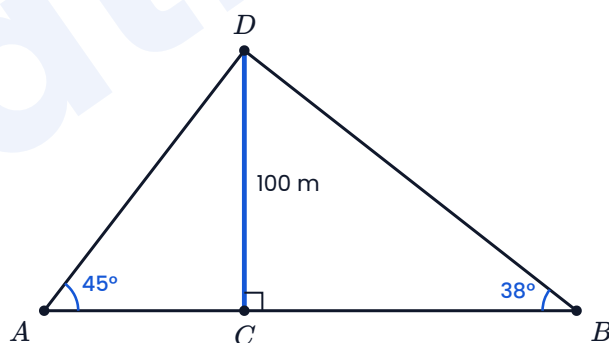
In $\triangle ACD$: $\tan 52^\circ = \frac{100}{AC}$. From the tables $\tan 52^\circ = 1.2799$, so $AC = \frac{100}{1.2799} \approx 78.13$ m.

In $\triangle BCD$: $\tan 45^\circ = \frac{100}{CB}$, so $CB = 100$ m.

$$AB = AC + CB = 78.13 + 100 = 178.13$$

Answer: $AB \approx 178$ m

3. From the top D of a tower CD 100 m high, two ships A and B on opposite sides of the tower are seen at angles of depression of 45° and 38° . The ships and the foot of the tower are in one straight line. Find the distance between the ships to the nearest metre. ICSE 2023, 4 marks



SOLUTION

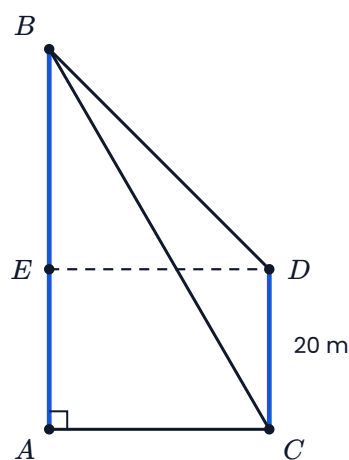
The angles of elevation of D from the ships equal the angles of depression (alternate angles).

$AC = \frac{100}{\tan 45^\circ} = 100$ m. From the tables $\tan 38^\circ = 0.7813$, so $CB = \frac{100}{0.7813} \approx 127.99$ m.

$$AB = 100 + 127.99 = 227.99$$

Answer: about 228 m

4. From the top of a cliff, the top and the bottom of a tower 20 m high are seen at angles of depression of 45° and 60° . Find (a) the height of the cliff (b) the distance between the cliff and the tower. **ICSE 2020, 4 marks**



SOLUTION

Let $AB = h$ be the cliff, $CD = 20$ m the tower and $AC = d$. Draw DE horizontal to meet the cliff at E .

From the bottom of the tower: $\tan 60^\circ = \frac{h}{d}$, so $h = \sqrt{3}d$.

From the top of the tower: $\tan 45^\circ = \frac{h - 20}{d}$, so $h - 20 = d$.

$\sqrt{3}d - 20 = d$, so $d = \frac{20}{\sqrt{3} - 1} = 10(\sqrt{3} + 1) \approx 27.32$ m and $h = d + 20 \approx 47.32$ m.

Answer: (a) about 47.32 m (b) about 27.32 m

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